PYTHON PROGRAMMING

Project Report

Gear Train Performance Analyzer

Calculation and visualisation of speed, torque and power of gear sets using Python

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1. Abstract

Gears are used in almost every mechanical system to change speed, increase torque or transmit power between shafts. Evaluating several gear sets by hand is slow and error-prone. This project presents a **Gear Train Performance Analyzer** written in Python that automates this task. The program reads the number of teeth of the driver and driven gears, the input speed, the input torque and the efficiency of each gear set from a CSV file, validates the data, and calculates the gear ratio, output speed, output torque, input power and output power. Each gear set is then classified as Excellent, Good, Average or Poor on the basis of its efficiency. The results are written to an output CSV file and visualised with five graphs created using Matplotlib. For the five gear sets analysed, the gear ratio ranged from 2.0 to 3.0, output speed from 500 to 750 RPM, and output torque from 74.4 to 115.2 Nm, with an average efficiency of 94 %. The project demonstrates the use of file handling, exception handling, loops, conditionals, and the pandas, NumPy and Matplotlib libraries for engineering data analysis.

Keywords: gear ratio, torque, output speed, power transmission, efficiency, pandas, Matplotlib.

2. Problem Statement and Objectives

Problem statement: Given the specifications of several gear sets, calculate and compare their output speed, output torque and power transmission, and present the findings in a clear numerical and graphical form, without performing manual calculations for every gear set.

Objectives

- To read gear data from a CSV file using Python.
- To validate the input data (teeth count and efficiency) before processing.
- To calculate gear ratio, output RPM, output torque, input power and output power for every gear set.
- To classify gear sets by performance level using their efficiency.

- To store the calculated results in an output CSV file.
- To generate graphs that make comparison between gear sets easy.
- To apply core Python programming concepts to a practical mechanical-engineering problem.

3. Introduction and Background

A gear is a toothed machine element that transmits motion and power from one shaft to another. In a simple gear pair, the smaller gear that receives power is called the **driver** (pinion) and the gear that is turned by it is the **driven** gear. The relationship between their sizes decides how speed and torque change from input to output.

Gear ratio. The gear ratio is the ratio of the number of teeth on the driven gear to that on the driver gear. A ratio greater than 1 reduces speed and multiplies torque (speed reducer); a ratio less than 1 does the opposite.

Power. Mechanical power in a rotating shaft depends on its speed and torque. A real gear set loses some power through friction, heat and churning of lubricant, so the output power is always lower than the input power. The ratio of output to input power is the efficiency.

Formulas used in this project

Quantity	Formula	Unit
Gear ratio (i)	$i = N_{driven} / N_{driver}$	-
Output speed	$n_{out} = n_{in} / i$	RPM
Output torque	$T_{out} = T_{in} \times i \times (eta / 100)$	Nm
Input power	$P_{in} = (2 \times pi \times n_{in} \times T_{in}) / 60$	W
Output power	$P_{out} = P_{in} \times (eta / 100)$	W

Here N is the number of teeth, n is rotational speed in RPM, T is torque in newton-metres and eta is efficiency in percent. Since $T_{out} \times omega_{out} = P_{in} \times eta/100$, the speed and torque formulas are consistent with the power formula: speed falls by the factor i while torque rises by the same factor, less the losses.

4. Methodology / Algorithm

The program follows a sequential, data-driven approach:

1. **Start** and print the program title.
2. **Check** that `gear_input.csv` exists; otherwise display an error and stop.
3. **Read** the CSV file into a pandas DataFrame (inside a try/except block) and display it.
4. **Verify** that the required columns DriverTeeth, DrivenTeeth, InputRPM, InputTorque and Efficiency are present.
5. **For each row** (gear set): (a) validate that teeth counts are positive and efficiency is in the range 0-100; skip invalid rows with a message; (b) compute gear ratio; (c) compute output speed; (d) compute output torque; (e) compute input power; (f) compute output power; (g) assign the performance rating.

6. **Add** the computed lists as new columns of the DataFrame and display the final table.
7. **Save** the table to `gear_results.csv`.
8. **Print** summary statistics (average ratio, max/min RPM, average and max torque, average efficiency).
9. **Generate** five graphs and save them in the `graphs` folder.
10. **Identify and print** the gear sets with highest speed, highest torque and best efficiency. **Stop.**

Performance rating rule

Efficiency (%)	Rating
>= 95	Excellent
>= 90 and < 95	Good
>= 80 and < 90	Average
< 80	Poor

Input dataset

Set	Driver Teeth	Driven Teeth	Input RPM	Input Torque (Nm)	Efficiency (%)
0	20	40	1200	50	95
1	25	50	1500	45	94
2	18	54	1800	35	92
3	30	60	1000	60	96
4	22	44	1400	40	93

5. Python Implementation and Programming Concepts Used

The project is implemented in a single Python 3 script, `main.py`. The complete code is given in Appendix A and in the GitHub repository.

Libraries

- **pandas** - reading and writing CSV files, storing data in a DataFrame, adding columns and finding maximum values.
- **NumPy** - the constant pi, and rounding of result arrays.
- **Matplotlib** - line plots, bar chart and histogram.
- **os** - checking whether files and folders exist and creating the `graphs` folder.

Programming concepts

Concept	Where it is used
File handling	`os.path.exists()`, `pd.read_csv()`, `to_csv()`
Exception handling	try/except around CSV reading
Variables and data types	Integers (teeth, RPM), floats (ratio, torque, power), strings (status)

Concept	Where it is used
Lists	gear_ratio, output_speed, output_torque, power_input, power_output, status
Loops	`for index, row in data.iterrows()`
Conditional statements	Data validation and if/elif/else performance classification
Arithmetic / math	Gear formulas, `np.pi`, `np.round()`
Data structures (DataFrame)	Adding columns, `idxmax()`, mean(), max(), min()
Data visualisation	`plt.plot()`, `plt.bar()`, `plt.hist()`, `plt.savefig()`
Defensive programming	Missing-file, missing-column and invalid-value checks

Core calculation block

```
for index, row in data.iterrows():
    driver_teeth = row["DriverTeeth"];  driven_teeth = row["DrivenTeeth"]
    input_rpm = row["InputRPM"];        input_torque = row["InputTorque"]
    efficiency = row["Efficiency"]

    if driver_teeth <= 0 or driven_teeth <= 0:      # validation
        print("Invalid Gear Teeth at Row", index + 1); continue
    if efficiency <= 0 or efficiency > 100:
        print("Invalid Efficiency at Row", index + 1); continue

    ratio  = driven_teeth / driver_teeth                        # gear ratio
    rpm    = input_rpm / ratio                                  # output speed
    torque = input_torque * ratio * (efficiency / 100)          # output torque
    pin    = (2 * np.pi * input_rpm * input_torque) / 60        # input power
    pout   = pin * (efficiency / 100)                           # output power

    if   efficiency >= 95: status.append("Excellent")
    elif efficiency >= 90: status.append("Good")
    elif efficiency >= 80: status.append("Average")
    else:                  status.append("Poor")
```

6. Results and Data Visualizations

6.1 Calculated results

Table 1 shows the output generated by the program (`gear_results.csv`).

Set	Ratio	Out RPM	Out Torque (Nm)	In Power (W)	Out Power (W)	Rating
0	2.0	600	95.0	6283.19	5969.03	Excellent
1	2.0	750	84.6	7068.58	6644.47	Good
2	3.0	600	96.6	6597.34	6069.56	Good
3	2.0	500	115.2	6283.19	6031.86	Excellent
4	2.0	700	74.4	5864.31	5453.80	Good

Table 1: Calculated performance of the five gear sets

6.2 Summary statistics

Parameter	Value
Average gear ratio	2.2
Maximum output RPM	750.0 (Gear set 1)

Parameter	Value
Minimum output RPM	500.0 (Gear set 3)
Average output torque	93.16 Nm
Maximum output torque	115.2 Nm (Gear set 3)
Average efficiency	94.0 %
Best efficiency	96 % (Gear set 3)

Table 2: Project summary generated by the program

6.3 Graphs

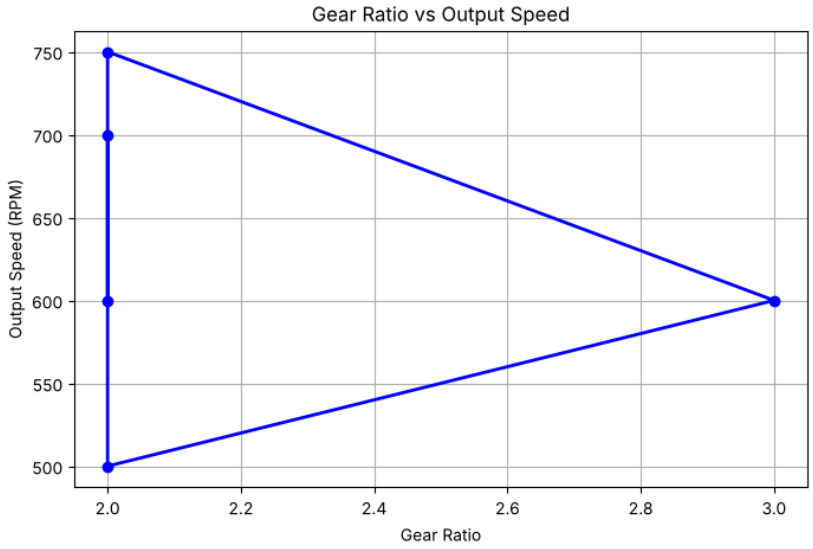


Figure 1: Gear ratio vs output speed

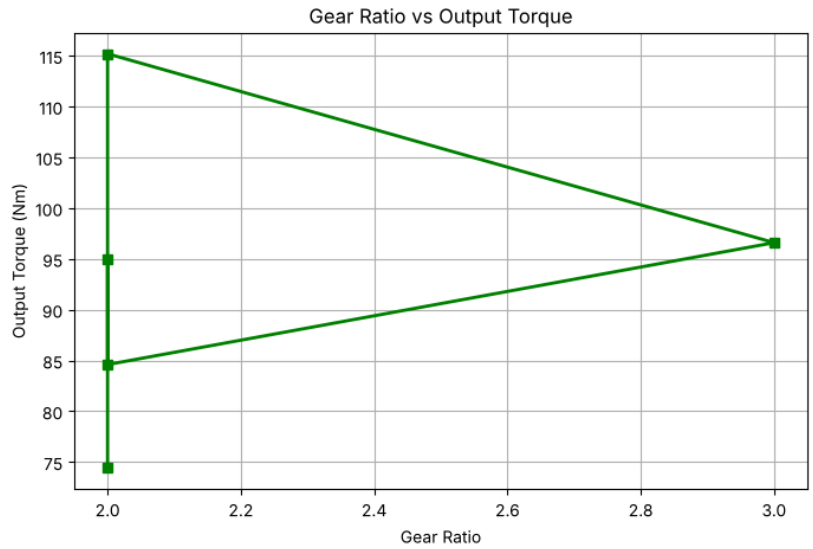


Figure 2: Gear ratio vs output torque

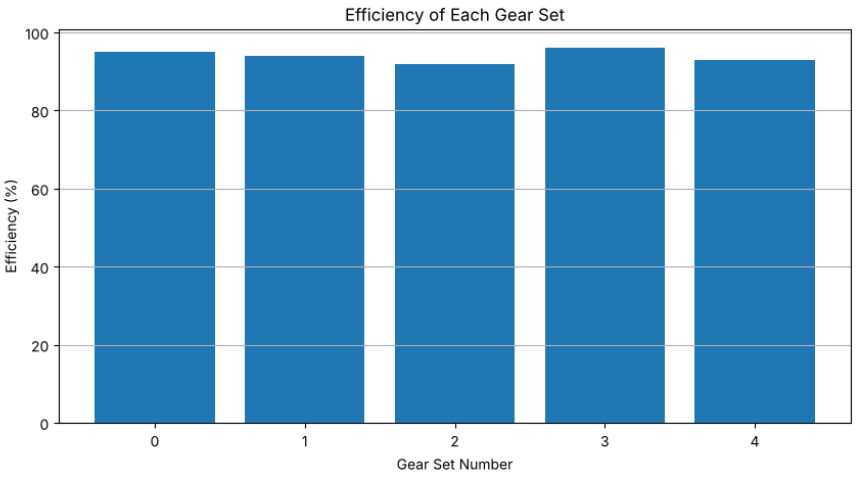


Figure 3: Efficiency of each gear set

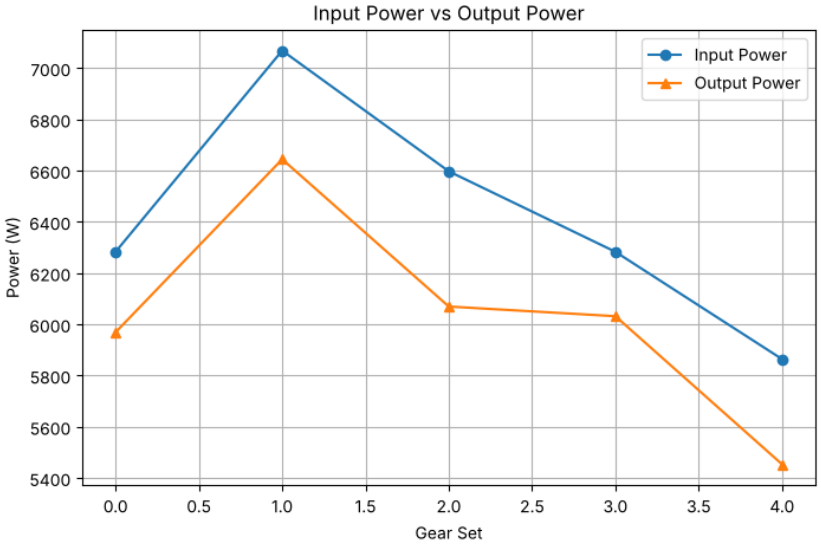


Figure 4: Input power vs output power

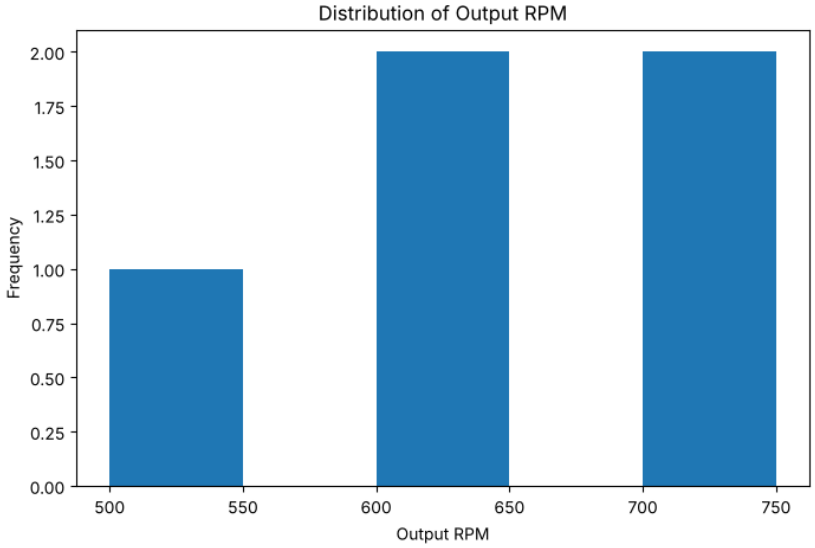


Figure 5: Distribution of output RPM

6.4 Program output (console)

Figure 6 shows the final-analysis part of the console output when the program is executed.

```
PROJECT SUMMARY
--------------------------------------------------
Average Gear Ratio      : 2.2
Maximum Output RPM      : 750.0
Minimum Output RPM      : 500.0
Average Output Torque   : 93.16
Maximum Output Torque   : 115.2
Average Efficiency      : 94.0 %
--------------------------------------------------

============================================================
FINAL ANALYSIS
============================================================

Highest Output Speed
--------------------
DriverTeeth            25
DrivenTeeth            50
InputRPM             1500
InputTorque            45
Efficiency             94
GearRatio             2.0
OutputRPM           750.0
OutputTorque         84.6
InputPower(W)     7068.58
OutputPower(W)    6644.47
Performance          Good
Name: 1, dtype: object
```

Figure 6: Console output - project summary and highest-speed gear set

7. Discussion and Observations

- **Effect of gear ratio.** Four of the five gear sets have a ratio of 2.0 and one (set 2, 18:54 teeth) has a ratio of 3.0. Every set is a speed reducer: output speed is lower than input speed and output torque is higher than input torque.
- **Set 3 gives the highest torque.** With a ratio of 2.0, an input torque of 60 Nm and the highest efficiency (96 %), it delivers 115.2 Nm but also the lowest output speed (500 RPM). This shows the speed-torque trade-off.
- **Set 1 gives the highest speed.** It has the highest input speed (1500 RPM) and a ratio of 2.0, so it outputs 750 RPM, but its output torque (84.6 Nm) is only the third-highest.
- **Set 2 (ratio 3.0)** halves speed more strongly: input 1800 RPM becomes 600 RPM, and the output torque of 96.6 Nm is the second highest even though its input torque is only 35 Nm, showing how a higher ratio multiplies torque.
- **Efficiency.** All efficiencies lie between 92 % and 96 %. Sets 0 and 3 are rated Excellent; sets 1, 2 and 4 are rated Good. None is Average or Poor.
- **Power loss.** The power lost in each set (input minus output) is shown in Table 3. Set 2 loses the most (527.78 W) because it has the lowest efficiency (92 %) at a high input power, while set 3 loses the least (251.33 W). Lost power appears as heat, so low-efficiency sets need better lubrication or cooling.

Set	Efficiency (%)	Input Power (W)	Output Power (W)	Power Loss (W)
0	95	6283.19	5969.03	314.16
1	94	7068.58	6644.47	424.11
2	92	6597.34	6069.56	527.78
3	96	6283.19	6031.86	251.33
4	93	5864.31	5453.80	410.51

Table 3: Power loss in each gear set

Limitations

- Figures 1 and 2 join the data points with straight lines in the order of the dataset. Because four sets share the same ratio, the lines run vertically and form a triangle shape. Since gear ratio is not a continuous time-like variable here, a scatter plot would represent these points more correctly.
- The dataset is small (five gear sets), so the statistics are only illustrative.
- The model uses a single efficiency value per gear set and ignores backlash, gear geometry, material strength, lubrication and load variation.
- Only a single-stage gear pair is analysed; multi-stage gear trains are not covered.

8. Conclusion

The Gear Train Performance Analyzer successfully automates the calculation of gear ratio, output speed, output torque and input/output power for multiple gear sets. The program validates its input, writes the results to a CSV file, and presents them through five graphs and a console summary. For the test data, the 30:60 gear set (set 3) gave the best efficiency (96 %) and the highest torque (115.2 Nm), while the 25:50 set (set 1) gave the highest output speed (750 RPM). The project confirms the basic principle that gears trade speed for torque, and that real gear sets always lose some power. It also gave practical experience with file handling, exception handling, loops, conditions, pandas, NumPy and Matplotlib.

Future scope

- Replace the line plots of Figures 1 and 2 with scatter plots, and add a gear-ratio vs power-loss plot.
- Extend the program to multi-stage gear trains and to helical, bevel and worm gears.
- Build a graphical user interface (Tkinter or Streamlit) for entering data interactively.
- Export a formatted report automatically (PDF or Excel).

9. References

[1] R. L. Norton, *Machine Design: An Integrated Approach*, Pearson.

[2] J. E. Shigley and C. R. Mischke, *Mechanical Engineering Design*, McGraw-Hill.

[3] R. S. Khurmi and J. K. Gupta, *Theory of Machines*, S. Chand Publishing.

[4] The pandas development team, pandas documentation. https://pandas.pydata.org/docs/

[5] NumPy documentation. https://numpy.org/doc/

[6] Matplotlib documentation. https://matplotlib.org/stable/

[7] Python Software Foundation, Python 3 documentation. https://docs.python.org/3/

[8] Project source code and dataset: https://github.com//gear-train-analyzer

Appendix A: Complete Source Code (main.py)

```
# ============================================================
# Gear Train Performance Analyzer
# Developed By : Aneesh Gopal (RA2311072010013)
# Language     : Python 3
# Libraries    : pandas, numpy, matplotlib
# ============================================================
# ------------------------------------------------------------
# Import Libraries
# ------------------------------------------------------------
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os
# ------------------------------------------------------------
# Program Title
# ------------------------------------------------------------
print("=" * 70)
print("        GEAR TRAIN PERFORMANCE ANALYZER")
print("=" * 70)
# ------------------------------------------------------------
# Check CSV File
# ------------------------------------------------------------
INPUT_FILE = "gear_input.csv"
OUTPUT_FILE = "gear_results.csv"
if not os.path.exists(INPUT_FILE):
    print("\nERROR : gear_input.csv not found.")
    print("Please keep the CSV file in the same folder.")
    exit()
# ------------------------------------------------------------
# Read CSV
# ------------------------------------------------------------
try:
    data = pd.read_csv(INPUT_FILE)
except Exception as e:
    print("Unable to read CSV File")
    print(e)
    exit()
# ------------------------------------------------------------
# Display Data
# ------------------------------------------------------------
print("\nINPUT DATA\n")
print(data)
print("\nNumber of Gear Sets :", len(data))
# ------------------------------------------------------------
# Required Columns
# ------------------------------------------------------------
required_columns = [
    "DriverTeeth",
    "DrivenTeeth",
    "InputRPM",
    "InputTorque",
    "Efficiency"
]
for column in required_columns:
    if column not in data.columns:
        print("\nMissing Column :", column)
        exit()
# ------------------------------------------------------------
# Empty Lists
# ------------------------------------------------------------
gear_ratio = []
output_speed = []
output_torque = []
power_input = []
power_output = []
status = []
print("\nStarting Calculations...\n")
# ============================================================
# Gear Calculations
# ============================================================
for index, row in data.iterrows():
    # Read Input Values
    driver_teeth = row["DriverTeeth"]
    driven_teeth = row["DrivenTeeth"]
    input_rpm = row["InputRPM"]
    input_torque = row["InputTorque"]
    efficiency = row["Efficiency"]
    # ---------------------------------------------------------
    # Validate Data
    # ---------------------------------------------------------
    if driver_teeth <= 0 or driven_teeth <= 0:
        print("Invalid Gear Teeth at Row", index + 1)
        continue
```

```
    if efficiency <= 0 or efficiency > 100:
        print("Invalid Efficiency at Row", index + 1)
        continue
    # ------------------------------------------------------------
    # Gear Ratio
    # ------------------------------------------------------------
    ratio = driven_teeth / driver_teeth
    gear_ratio.append(ratio)
    # ------------------------------------------------------------
    # Output Speed
    # ------------------------------------------------------------
    rpm = input_rpm / ratio
    output_speed.append(rpm)
    # ------------------------------------------------------------
    # Output Torque
    # ------------------------------------------------------------
    torque = input_torque * ratio * (efficiency / 100)
    output_torque.append(torque)
    # ------------------------------------------------------------
    # Input Power
    #
    # P = (2πNT)/60
    # ------------------------------------------------------------
    pin = (2 * np.pi * input_rpm * input_torque) / 60
    power_input.append(pin)
    # ------------------------------------------------------------
    # Output Power
    # ------------------------------------------------------------
    pout = pin * (efficiency / 100)
    power_output.append(pout)
    # ------------------------------------------------------------
    # Performance Status
    # ------------------------------------------------------------
    if efficiency >= 95:
        status.append("Excellent")
    elif efficiency >= 90:
        status.append("Good")
    elif efficiency >= 80:
        status.append("Average")
    else:
        status.append("Poor")
# ============================================================
# Add Results into DataFrame
# ============================================================
data["GearRatio"] = np.round(gear_ratio, 3)
data["OutputRPM"] = np.round(output_speed, 2)
data["OutputTorque"] = np.round(output_torque, 2)
data["InputPower(W)"] = np.round(power_input, 2)
data["OutputPower(W)"] = np.round(power_output, 2)
data["Performance"] = status
# ============================================================
# Display Results
# ============================================================
print("=" * 80)
print("FINAL RESULTS")
print("=" * 80)
print(data)
# ============================================================
# Save Results
# ============================================================
data.to_csv(OUTPUT_FILE, index=False)
print("\nResults saved successfully as:", OUTPUT_FILE)
# ============================================================
# Project Statistics
# ============================================================
print("\nPROJECT SUMMARY")
print("-" * 50)
print("Average Gear Ratio      :", round(data["GearRatio"].mean(), 2))
print("Maximum Output RPM      :", round(data["OutputRPM"].max(), 2))
print("Minimum Output RPM      :", round(data["OutputRPM"].min(), 2))
print("Average Output Torque   :", round(data["OutputTorque"].mean(), 2))
print("Maximum Output Torque   :", round(data["OutputTorque"].max(), 2))
print("Average Efficiency      :", round(data["Efficiency"].mean(), 2), "%")
print("-" * 50)
# ============================================================
# GRAPH GENERATION
# ============================================================
# Create folder for graphs if it does not exist
if not os.path.exists("graphs"):
    os.makedirs("graphs")
# ------------------------------------------------------------
# Graph 1 : Gear Ratio vs Output Speed
# ------------------------------------------------------------
plt.figure(figsize=(8,5))
plt.plot(
    data["GearRatio"],
    data["OutputRPM"],
```

```
    marker="o",
    linewidth=2,
    color="blue"
)
plt.title("Gear Ratio vs Output Speed")
plt.xlabel("Gear Ratio")
plt.ylabel("Output Speed (RPM)")
plt.grid(True)
plt.savefig("graphs/speed_vs_ratio.png")
# ------------------------------------------------------------
# Graph 2 : Gear Ratio vs Output Torque
# ------------------------------------------------------------
plt.figure(figsize=(8,5))
plt.plot(
    data["GearRatio"],
    data["OutputTorque"],
    marker="s",
    linewidth=2,
    color="green"
)
plt.title("Gear Ratio vs Output Torque")
plt.xlabel("Gear Ratio")
plt.ylabel("Output Torque (Nm)")
plt.grid(True)
plt.savefig("graphs/torque_vs_ratio.png")
# ------------------------------------------------------------
# Graph 3 : Efficiency Comparison
# ------------------------------------------------------------
plt.figure(figsize=(10,5))
plt.bar(
    range(len(data)),
    data["Efficiency"]
)
plt.title("Efficiency of Each Gear Set")
plt.xlabel("Gear Set Number")
plt.ylabel("Efficiency (%)")
plt.grid(axis="y")
plt.savefig("graphs/efficiency.png")
# ------------------------------------------------------------
# Graph 4 : Input Power vs Output Power
# ------------------------------------------------------------
plt.figure(figsize=(8,5))
plt.plot(
    data["InputPower(W)"],
    marker="o",
    label="Input Power"
)
plt.plot(
    data["OutputPower(W)"],
    marker="^",
    label="Output Power"
)
plt.title("Input Power vs Output Power")
plt.xlabel("Gear Set")
plt.ylabel("Power (W)")
plt.legend()
plt.grid(True)
plt.savefig("graphs/power_comparison.png")
# ------------------------------------------------------------
# Graph 5 : Output Speed Distribution
# ------------------------------------------------------------
plt.figure(figsize=(8,5))
plt.hist(
    data["OutputRPM"],
    bins=5
)
plt.title("Distribution of Output RPM")
plt.xlabel("Output RPM")
plt.ylabel("Frequency")
plt.savefig("graphs/output_speed_histogram.png")
# ------------------------------------------------------------
# Display all graphs
# ------------------------------------------------------------
plt.show()
# ============================================================
# FINAL ANALYSIS
# ============================================================
print("\n" + "="*60)
print("FINAL ANALYSIS")
print("="*60)
highest_speed = data.loc[data["OutputRPM"].idxmax()]
highest_torque = data.loc[data["OutputTorque"].idxmax()]
best_efficiency = data.loc[data["Efficiency"].idxmax()]
print("\nHighest Output Speed")
print("--------------------")
print(highest_speed)
print("\nHighest Output Torque")
```

```
print("---------------------")
print(highest_torque)
print("\nBest Efficiency")
print("----------------")
print(best_efficiency)
print("\nGraphs saved successfully inside the 'graphs' folder.")
print("Analysis completed successfully.")
```

Appendix B: Complete Program Output (console)

```
======================================================================
        GEAR TRAIN PERFORMANCE ANALYZER
======================================================================

INPUT DATA

   DriverTeeth  DrivenTeeth  InputRPM  InputTorque  Efficiency
0           20           40      1200           50          95
1           25           50      1500           45          94
2           18           54      1800           35          92
3           30           60      1000           60          96
4           22           44      1400           40          93

Number of Gear Sets : 5

Starting Calculations...

================================================================================
FINAL RESULTS
================================================================================
   DriverTeeth  DrivenTeeth  ...  OutputPower(W)  Performance
0           20           40  ...         5969.03    Excellent
1           25           50  ...         6644.47         Good
2           18           54  ...         6069.56         Good
3           30           60  ...         6031.86    Excellent
4           22           44  ...         5453.80         Good

[5 rows x 11 columns]

Results saved successfully as: gear_results.csv

PROJECT SUMMARY
--------------------------------------------------
Average Gear Ratio      : 2.2
Maximum Output RPM      : 750.0
Minimum Output RPM      : 500.0
Average Output Torque   : 93.16
Maximum Output Torque   : 115.2
Average Efficiency      : 94.0 %
--------------------------------------------------

============================================================
FINAL ANALYSIS
============================================================

Highest Output Speed
--------------------
DriverTeeth            25
DrivenTeeth            50
InputRPM             1500
InputTorque            45
Efficiency             94
GearRatio             2.0
OutputRPM           750.0
OutputTorque         84.6
InputPower(W)     7068.58
OutputPower(W)    6644.47
Performance          Good
Name: 1, dtype: object

Highest Output Torque
---------------------
DriverTeeth              30
DrivenTeeth              60
InputRPM               1000
InputTorque              60
Efficiency               96
GearRatio               2.0
OutputRPM             500.0
OutputTorque          115.2
InputPower(W)       6283.19
OutputPower(W)      6031.86
Performance       Excellent
Name: 3, dtype: object

Best Efficiency
---------------
DriverTeeth              30
```

```
DrivenTeeth              60
InputRPM               1000
InputTorque              60
Efficiency               96
GearRatio               2.0
OutputRPM             500.0
OutputTorque          115.2
InputPower(W)       6283.19
OutputPower(W)      6031.86
Performance       Excellent
Name: 3, dtype: object

Graphs saved successfully inside the 'graphs' folder.
Analysis completed successfully.
```

Full results table (gear_results.csv)

	DriverTeeth	DrivenTeeth	InputRPM	InputTorque	Efficiency	GearRatio	OutputRPM	OutputTorque	InputPower(W)	OutputPower(W)	Performa
0	20	40	1200	50	95	2.0	600.0	95.0	6283.19	5969.03	Excelle
1	25	50	1500	45	94	2.0	750.0	84.6	7068.58	6644.47	G
2	18	54	1800	35	92	3.0	600.0	96.6	6597.34	6069.56	G
3	30	60	1000	60	96	2.0	500.0	115.2	6283.19	6031.86	Excelle
4	22	44	1400	40	93	2.0	700.0	74.4	5864.31	5453.80	G